Task #3 : The overfitting Astronaut

My rule :

looking only at the 6 photos, the features they all shares are a blue shirt, a white sign, black follow text that is roughly upright, and daylight. So the rule I would honestly learn is if the person wears a blue shirt and holds a white sign with follow in black marker, upright and it is daylight/bright. otherwise No.

Applying my rule :

A : white EVA suit, red Follow, bright sun.

Rule check : no blue shirt, text is red.

Answer : NO.

Truth : follow

Result : Wrong.

B. Blue shirt, follow upside down, cloudy

Rule check: text is not upright, not bright

ans: NO

Truth: follow

Result: Wrong.

C. blue shirt, Stop in black, bright sun.

Rule check: the word is not follow

ans: NO

Truth: Not follow

Result: Correct

D. White EVA suit, hand-dreawn arrow, bright sun

Rule check: No blue shirt, no word.

ans

~~Truth~~: NO

Truth: follow

Result: Wrong

Reflection Answer

① 1 out of 4. It failed on A, B and D, the three ~~real~~ real follow signs, so it is ~~a~~ a model that rejects everything slightly unfamiliar. The one correct answer is weak evidence, a sloppier rule like "blue shirt + bright sun" would say yes to C and score 0/4.

② the rule locked onto the shirt colour, which happened to be constant in the training photos but it's not what defines a follow sign. This is data bias. The traning set never showed a white suit or a different marker colour, so the model learned "blue shirt" = follow instead of "this sign" = follow. D also shows the sign itself can be a team chosen image, so trying the rule to the word follow was too narrow.

(3) All six training photos had upright text in bright sun, so the rule memorised those exact conditions. An upside down sign on a cloudy day is the same concept in a new condition. This is overfitting. memorising incidental details of the examples instead of the general concept

(4) (a) White EVA suit holding a sign with follow in red marker. This breaks the "blue shirt" and "black ink" correlations.

(b) A different person and suit, sign upside down or rotated, in cloudy or low light. This breaks the orientation and lighting correlations.

(c) A blue shirted person holding a stop sign or a blank board, labelled "not follow". Every current photo is a positive, so the model has never seen what a non-follow sign looks like. This is the gap in the original set.

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(5) The rover would learn "blue shirt = follow". It would miss the astronaut in a white EVA suit. and might wrongly react to any blue-clothed person or blue object near the course. That's failed task and a safety risk. A better training set would:

1. vary the suit, marker colour and sign design.
2. vary lighting, background, distance, tilt, rotation, and partial occlusion.
3. include many negatives.
4. Add augmentation
5. keep a held-out test set of conditions never trained on, and score each condition separately instead of one overall number.

A good model must key on the sign itself and ignore the shirt, lighting and orientation. A training set with no variety and no negatives only teaches shortcuts.

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